

Cerisse Code Modifications Overview

Branch `ibm_gpu` vs. `eib_dev` baseline

Auto-generated changelog

March 2026

1 Summary

This document catalogues *all* modifications on the `ibm_gpu` branch relative to the `eib_dev` baseline commit (8666588). Changes fall into three categories:

1. **GPU correctness & portability** — fixes for data races, CUDA device-code issues, and lambda capture bugs.
2. **RZ axisymmetric coordinate support** — new finite-volume divergence, geometric sources, volume-weighted diagnostics.
3. **IBM (Immersed Boundary Method) refactor** — GPStore, BVH acceleration, modular file split, moving-geometry support.

2 File-by-File Change Table

File	Status	Description
<i>Core solver (non-IBM)</i>		
<code>src/CNS.h</code>	kept	Added <code>rz_sanity_check()</code> static method; checks <code>SPACEDIM==2</code> , non-periodic r , and <code>prob_lo[0]==0</code> . Added <code>#include <AMReX_Math.H></code> .
<code>src/CNS.cpp</code>	kept	maxEigen(): Rewritten from <code>AsyncArray</code> to <code>ReduceOps<ReduceOpMax></code> ; original had a GPU data race. Per- <code>SPACEDIM</code> specialisation (1D/2D/3D).
	kept	printTotal(): Volume-weighted integral via <code>geom.GetVolume()</code> + <code>ReduceOps<ReduceOpSum></code> ; required for correct RZ integrals.
	kept	postCoarseTimeStep(): Commented out the <code>computeSURFs</code> + <code>FillPatch</code> path; GPStore refactor has not yet re-enabled surface force computation at coarse-time-step.
	kept	Constructor calls <code>rz_sanity_check(Geom())</code> .

File	Status	Description
src/cls/Thermodynamics.h	kept	Replaced <code>std::max(min_euler_press, P)</code> with <code>if (P < ...) P = ...</code> in <code>ensurePTYfillq</code> (both <code>calorifically_perfect_gas_t</code> and <code>multispecies_pele_gas_t</code>). Reason: <code>std::max</code> ODR-uses the <code>constexpr</code> by reference, undefined in CUDA device code.
src/rhs/Weno.h	kept	Changed <code>[=,this]</code> $\rightarrow$ <code>[=]</code> in GPU lambda inside <code>ParallelFor</code> ; capturing <code>this</code> is invalid in device code.
<i>Time integration</i>		
src/tim/advance.cpp	kept	Moving geometry: New <code>#ifdef AMREX_USE_GPIBM</code> block calls <code>update_geometry()</code> $\rightarrow$ <code>rebuildGeometryData()</code> $\rightarrow$ <code>rebuildIBM()</code> at each step when <code>ib_move</code> is true.
	kept	GPStore 3-phase: Phase 1: level-wide <code>cons2prims</code> into <code>prims_mf</code> . Phase 2: <code>computeAllGPs(prims_mf)</code> . Phase 3: <code>prims2cons</code> at GP cells only. Replaces per-FAB <code>computeGPs</code> .
	kept	Changed <code>cls.h.prims2cons</code> $\rightarrow$ <code>cls.d->prims2cons</code> and loop bound to <code>PROB::ProbClosures::NCONS</code> for GPU safety.
src/tim/compute_rhs.cpp	kept	Compile guard: <code>#error</code> if both <code>AMREX_USE_GPIBM</code> and <code>CNS_USE_EB</code> are defined (marker paths incompatible).
	kept	GPStore pre-loop: Level-wide <code>cons2prims</code> + <code>computeAllGPs</code> before the <code>MFIter</code> loop; inside the loop <code>prims</code> aliases <code>prims_mf</code> .
	kept	geoMarkers: Direct alias to <code>IBMMultiFab</code> / <code>EBMultiFab</code> array; removes the old auxiliary <code>BaseFab<uint8_t></code> + <code>ParallelFor</code> copy.
	kept	RZ divergence: Geometry-aware FV divergence with cylindrical metrics; <code>p/r</code> Euler source; viscous geometric source call. See <code>docs/changes_rz.tex</code> .
	kept	Solid RHS zeroing: Replaced <code>state *= (1-marker)</code> with explicit <code>if (marker != 0)</code> branch + inner loop. Avoids floating-point multiplication for a boolean mask.
<i>IBM module — modified files</i>		
src/ibm/IBMMultiFab.h	kept	Rewritten: cleaner doxygen, explicit <code>delete</code> copy ctors, <code>default</code> move ctors; <code>copytoRealMF()</code> helper. Forces <code>TheManagedArena()</code> .
src/ibm/eib.h	kept	Major refactor (-2157 / $+1030$ lines): <code>eib_t</code> template now includes modular headers; adds <code>GPStore</code> , <code>computeAllGPs()</code> , <code>rebuildGeometryData()</code> .
src/ibm/eib_cgal.h	kept	Extended CGAL wrappers; new BVH-based inside/outside testing and closest-point queries.

File	Status	Description
src/ibm/ib_walltypes.h	kept	Updated wall-model parameter structs, added interior_is_solid flag, multi-arg compute_surfIB.
src/ibm/Make.package	kept	Updated include paths for new header split.
<i>IBM module — new files</i>		
src/ibm/bvh.types.h	new	POD types: Point, AABB, BVHNode, TriMesh, Polygon2D, SurfElem, LocalFrame (478 lines).
src/ibm/bvh.h	new	BVH tree: Morton-code build, stack-based GPU closest-point traversal for TriMesh/Polygon2D (524 lines).
src/ibm/eib_bvh.h	new	EIB-BVH bridge: InsideTester (ray-cast majority vote), STL/OFF/2D-polygon reader, geometry-cache builder (942 lines).
src/ibm/eib_data.h	new	SoA data: gpData_t, surfImp_t, surfPhys_t, FaceCSR, compile-time constants (637 lines).
src/ibm/eib_interp.h	new	Interpolation helpers: image-point search, bi-/tri-linear weights, Lagrange extrapolation, local/global frame transforms (519 lines).
src/ibm/eib_io.h	new	I/O helpers: read_geom(), VTK surface output (.vtp), MPI gather, CSR builder (544 lines).

3 Reverted Changes

During a cleanup pass the following modifications were identified as debug/test artifacts and reverted to the eib_dev baseline:

Item	Reason for revert
const int istep in compute_rhs.cpp	Only used by the diagnostic print block (also removed).
50-line /* ... */ diagnostic block in RZ divergence	Debug AllPrint profiling of RZ-vs-Cartesian RHS; not needed in production.
Commented-out rhs_cartesian + unused inv_dr	Leftover from RZ development comparison.
avgDown() closing brace indentation in staged CNS.cpp	Whitespace-only bug introduced by staging.